

WILMS

Women's Interest-Free Loan Management System

Product Dossier — v1.8.0

WILMS Product Dossier

Version: 1.7.3 Classification: Confidential — Executive / Partner Distribution

Executive Summary

WILMS (Women's Interest-Free Loan Management System) is a production-grade operational platform for managing women's interest-free group lending programmes. Deployed on Vercel with Neon PostgreSQL, WILMS covers the complete lending lifecycle with strong RBAC, financial integrity controls, and board-ready reporting.

Platform features through v1.7.2. Documentation library through v1.7.3.

Product Overview

Interest-free loan management covering registration, approval, disbursement, weekly collections, reconciliation, expenses, communications, executive intelligence, and audit — designed for NGOs, government programmes, and institutional partners in Ghana and beyond.

Problem Statement

Paper and spreadsheet programmes lack separation of duties, auditability, and timely portfolio visibility. Field-to-HQ cash reconciliation gaps create fraud risk. Leadership lacks real-time KPIs for decision-making.

Solution

Modular TypeScript monolith on Next.js + Vercel + Neon with RBAC, operational pool ledgers, multi-channel notifications, executive intelligence, and contextual export capabilities.

Target Users

Collectors, registration officers, approvers, auditors, super admins, programme directors, NGO boards, procurement committees, and institutional investors.

Key Features

- Borrower lifecycle with document capture and GPS
- Loan approval with maker-checker controls
- Capital pool management with hard-stops
- Field collections with offline support
- Daily reconciliation and overpayment review
- Expense tracking with separation of duties

- Executive intelligence and forecasting
- Multi-channel notifications
- Contextual exports (PDF, Excel, CSV, Print)
- Operations incidents and maintenance windows

Architecture

Next.js 14 App Router hosts @wilms/domain via Route Handlers. Custom HMAC session authentication. Neon PostgreSQL with Drizzle ORM. Optional Redis rate limits.

Financial Model

Integer pesewas money. Pool replenishment, disbursement, repayment, adjustment ledger. Expenses affect operating cash only. Not a statutory GL.

Security Model

HMAC sessions, RBAC with five roles, CSRF protection, audit logging, upload allowlists, rate limiting, confidentiality notices on exports.

Deployment

Vercel production + Neon. Node.js 22.x. Migrations via domain SQL journal. Demo users for evaluation.

Certification Status

Production operational platform. Financial integrity controls verified. Statutory GL and multi-org deferred.

Roadmap Summary

v1.8 Integrations & Payments !' v1.9 Enterprise Automation !' v2.0 GL & Multi-b

v1.8.0 market-readiness (this sprint): guarantor capacity of three, geographic review cascade, official document IDs, branded exports, group-collection GPS exceptions, collector update requests, and mandatory borrower SMS.

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